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Successful Fast Track Implementation of Temporary Stimulation Vessel for Deepwater Acid Job Campaign

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Abstract

In this paper we detail Operator's strategic response to an unforeseen delay of a purpose-built stimulation vessel by converting a platform supply vessel (PSV) into a temporary stimulation vessel. We highlight the solutions and strategic collaborations necessary to overcome significant challenges and achieve operational objectives in a remote and resource-constrained location for deepwater acid job stimulation campaign of 2000 bbl of HCl 15% per well.

The approach involved converting a PSV into a fully equipped temporary stimulation vessel using skid-mounted pumping equipment. The process began with early engagement with class society authority to meet regulatory requirements. Daily reviews were held by Operator involving all stakeholders, including the shipyard, marine design company, contractor, and class society authority, to ensure seamless coordination. Key steps encompassed meticulous planning, efficient resource allocation, and early reviews by the class authority, addressing logistical challenges, installing equipment, and commissioning within an expedited timeline.

The project team achieved operational readiness in approximately 3 months, successfully minimizing the impact of the delay. The temporary setup used the same equipment set positioned on the rig deck, leading to the successful completion of the acid stimulation campaign. Comprehensive steps were undertaken to meet regulatory requirements, including necessary modifications and safety equipment installations. The strategic decisions to logistical challenges underscore the importance of flexibility, regulatory engagement, and rapid problem-solving in offshore project management. The converted stimulation vessel met the desired stimulation objectives, improved turnaround per job, and enhanced controls over health, safety, and environmental hazards and rig time. The project's success in a remote and resource-constrained location demonstrates the effectiveness of the approach and provides valuable insights for future projects facing similar challenges.

The paper presents the ability to design, build, and deploy a fit-for-purpose stimulation vessel in remote and resource-constrained locations within a limited time frame. The operator was able to quickly deploy a project team and identify and mobilize the resources available across the region. In the meantime, we

obtained approval and exemption from the class society to temporary set up the platform supply vessel for the entire stimulation campaign.

Introduction

In the offshore oil and gas industry, unexpected delays can significantly impact project timelines and operational efficiency. In this paper, we present the case of an operator's strategic response to an unforeseen delay in the availability of a purpose-built stimulation vessel for an acid stimulation campaign. The campaign encountered a critical challenge when mandatory ship maintenance activities for the purpose-built stimulation vessel extended beyond the project start date. Given the constraints of the completion schedule, the operator collaborated with its contractor to devise an alternative solution involving the conversion of a Platform Supply Vessel into a temporary, fully equipped stimulation vessel using skid-mounted pumping equipment within an expedited timeline. Through meticulous planning, efficient resource allocation, and strategic collaboration with all stakeholders, the project team achieved operational readiness in about three months. The project's success in such a remote and resource-constrained location highlights fit for purpose solutions and strategic collaborations necessary to overcome significant logistical challenges and achieve operational objectives in a short period.

In this paper, we will focus on the comprehensive steps undertaken by the stakeholders to meet flag regulatory and class requirements. It also outlines the main modifications, safety equipment installations, and the main challenges addressed throughout the process. This tailored approach facilitated the approval of key requirements for the well stimulation vessel conversion and enabled the flag authority to grant a dispensation to operate within a specific period under specific conditions included in the reviewed documentation. This strategic response underscores the importance of flexibility, regulatory engagement, and rapid problem-solving in offshore project management. The converted stimulation vessel achieved the desired stimulation objectives and led to improved turnaround per job, better controls over HSE hazards, and saved about 2 rig days per operation.

Acid stimulation represents a strategic avenue to maximize the extraction of hydrocarbons from deepwater reservoirs. By leveraging advanced acid stimulation techniques, the operator aims to enhance the productivity of their offshore wells, ensuring a more efficient and sustainable recovery of oil and gas. The process involves injecting specialized acid blends into the formation to enhance the productivity of oil facilitating the flow of hydrocarbons.

One of the primary challenges of matrix acid stimulation in offshore environments is the substantial volume of acid required. Offshore wells in carbonate reservoir are usually drilled with long horizontal sections, necessitating a higher volume of acid to achieve effective stimulation. Transporting and handling such large volumes of acid on drilling ship, which have limited space and storage capacity, poses significant logistical challenges.

In a bid to optimize their offshore acid stimulation operations for the 2025 campaign, the operator has embarked on fit for purpose project to convert a Platform Supply Vessel (PSV) (Figure 1) into a dedicated stimulation vessel. This conversion involves outfitting the PSV with state-of-the-art acid handling and injection systems, as well as advanced monitoring and control technologies. The transformed vessel will serve as a mobile stimulation unit, capable of executing acid treatments with precision and efficiency across various offshore wells.

The PSV's robust design and ample deck space provide an ideal platform for accommodating the specialized equipment required for acid stimulation operations. The vessel can be equipped with high-pressure pumps, acid storage tanks, and mixing units, ensuring seamless preparation and delivery of acid treatments. Additionally, the conversion enhances the vessel's versatility, enabling it to perform a range of offshore support functions beyond stimulation, such as logistics and supply chain management.

Figure 1—Platform Supply Vessel (PSV)

Compared to a scenario of acid job from Drilling Ship, the deployment of a dedicated stimulation vessel significantly reduces the logistical challenges associated with deepwater operations. The vessel's mobility allows for rapid response to stimulation needs across different well sites, minimizing downtime and maximizing operational efficiency. This approach also enhances safety by centralizing acid handling and injection processes thereby reducing the risk of accidents and environmental incidents.

Background

The operator planned to complete and stimulate four wells in H2 2024 in its field. The project began with a pressing need for a stimulation vessel to support their operational activities. The regionally suitable stimulation vessel, deemed suitable for the project, was still engaged in prior commitments, thereby introducing potential risks to the operator's schedule. This critical situation required immediate action. The requirement to comply with stringent class standards further complicated the task.

Vessel Selection

The process of selecting and customizing the Platform Supply Vessel (PSV) for conversion into a dedicated stimulation vessel requires a thorough and methodical approach and daily review meeting. The primary objectives were to ensure the vessel's suitability for acid stimulation scope and to meet all regulatory and operational requirements.

The selection of the PSV was based on several key criteria:

- Deck Space and Layout: Ample deck space was essential to accommodate the designed acid mixing, storage, and pumping equipment, targeting the scope of work involving 2000 bbl of 15% HCl pumped at a maximum rate of 8.5 bpm from the vessel to the rig. This also included space for the acquisition control cabin, air compressor, emergency quick disconnect, and high- pressure hose. Additionally, sufficient space between equipment was necessary to rig up the pipes and ensure safe movement for personnel.

- Structural Integrity: The vessel required robust structural features to withstand the high-pressure operations associated with acid stimulation.
- Versatility: The ability of the vessel to perform supplementary offshore support roles, including logistics and supply chain management in particular for bulk material, was considered an asset.
- Availability of Bed Space: Given the extended nature of offshore operations, adequate bed space count was needed for the day and the night shifts of 10 stimulation personnel per shift to operate the job, in addition to the vessel crew.
- Dynamic Positioning (DP2) Capability: The vessel needed to be equipped with a DP2 system for precise station-keeping during stimulation operations, crucial for maintaining the vessel's position relative to the rig in challenging offshore environments, thus enhancing safety and efficiency.
- Underdeck Tank: Availability of an underdeck tank to store freshwater, brine, and base oil was essential. This storage capability was vital for the acid stimulation process, enabling efficient and uninterrupted operations while ensuring the tank connection to main deck equipment did not compromise stability and safety.

Beside the selection of the vessel, it was critical to ensure there was a facility in the country able to carry out such conversion with adequate resources. The shipyard was identified and engaged at the early stage to be part of the project. Among the available vessel, one of the PSV in Operator's fleet met the selection criteria (Figure 1). Moreover, the vessel just completed dry dock maintenance and class inspection on the marine side.

Once the appropriate vessel was selected, the customization process involved several critical steps:

Design stage

The design phase started with setting up the project team that involved the operator, the stimulation contractor, the vessel owner, the marine design company and the shipyard. A comprehensive review of class requirements was completed, and a Gantt chart was built targeting operational objectives. The design phase of each section included in the following steps, started with a clear understanding of the stimulation operation requirement and the work environment. Then it was essential to identify all equipment fit for the operation and then elaborate on the general arrangement.

Final Equipment List

Prior to elaborate the stowage plan, the equipment list was finalized to enable a treatment of 2000bbl of 15% HCl (1.8-8.5 bpm) and footprint discussed and agreed. The equipment list includes:

- Acid Storage Tanks: High-capacity, acid-resistant tanks for storing large volumes of acid. 9 acids storage tanks of 131 bbl were airfreighted in the country for this purpose. 4 additional tanks of the same capacity were mobilized by sea to serve as back-up.
- A blending unit of 2 × 50 bbl stainless steel tanks for precise batch mixing of acid treatment. Two centrifugal pumps to boost the high-pressure pumps with fluids.
- Two high-pressure pumps of total 3000 HHP for the controlled delivery of acid working in redundancy. The pumps rigged up to the 2 in high pressure line enabled to pump at a rate between 1.8 bpm to 8.5 bpm up to 10000 psi.
- A control cabin: for operational control of the pumps and real time data acquisition and monitoring of treatment. The controls system includes surface treating pressure and pumping rate data collected from existing devices in the rig up and equipment.
- An air compressor for air supply for air driven equipment. Vessel air supply system considered as back up.

- Safety Equipment: Personal protective equipment (PPE), spill containment kits, and emergency showers. Foam and dry chemical fire suppression systems for hazardous areas.

Engineering phase

This process is further complemented by the thorough review of the vessel's structural, electrical, piping, and safety drawing details. This review is crucial to evaluate the need for reinforcement to accommodate the additional weight and the stimulation operational needs. During the engineering phase, the following documents have been prepared and submitted for Class approval:

- Stowage layout plan
- Sea-fastening and structural grillage calculation & arrangement drawing, with including design analysis and drawing of the additional gooseneck
- Hazardous area plan

All the documents (structural analysis, plan, P&ID,..) have been than distributed to Class for approval with additional vessel operations manual and fire & safety plan dedicated for acid job execution.

Stowage Plan. The design and finalization of the stowage plan (Figure 1Figure 2) was dynamic, as the preliminary draft stowage plan had to be reviewed at multiple steps of the design to ensure equipment positioning meets structural constraints, hazards area limitations, piping and vent requirement and all existing vessel safety measures:

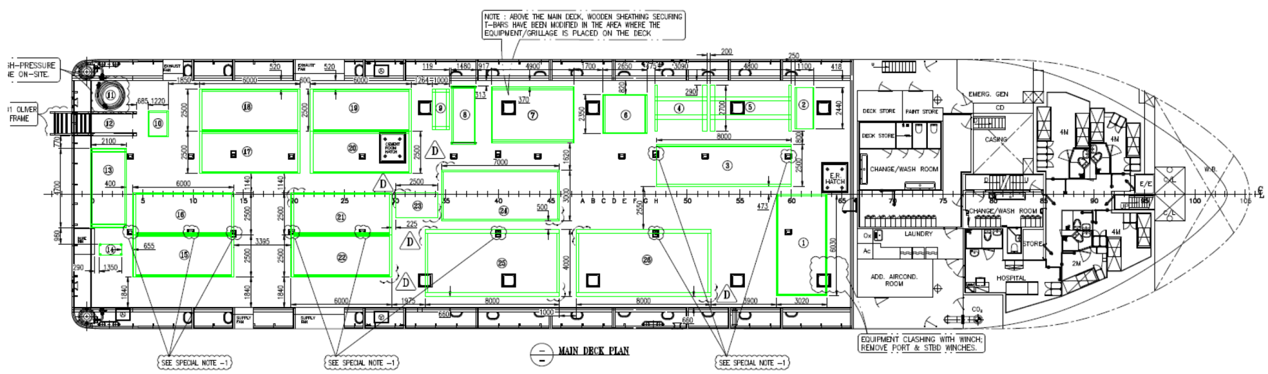

Figure 2—Stowage Plan

- Equipment Placement: Strategic placement of equipment to avoid congestion and facilitate operations.
- Securing Methods: Use of securing methods with brackets to prevent movement during transit.
- Accessibility: Ensuring all equipment is easily accessible for maintenance and emergency response.
- Additionally, it was crucial to ensure that the acid tanks are above the acceptable minimum distance from accommodation, control station and access doors,

Sea fastening and grillage plan. Stability and deck loading analysis was then conducted to ensure the vessel could handle dynamic forces during operation and support the loading of the equipment per square meter. During the design phase, a grillage was required to distribute loads evenly for some equipment. In addition to the grillage, it was also required to design and fabricate a gooseneck (Figure 4) to prevent the hose from exceeding the Minimum Bend Radius (MBR) and to withstand the weight of the high-pressure hose under operation.

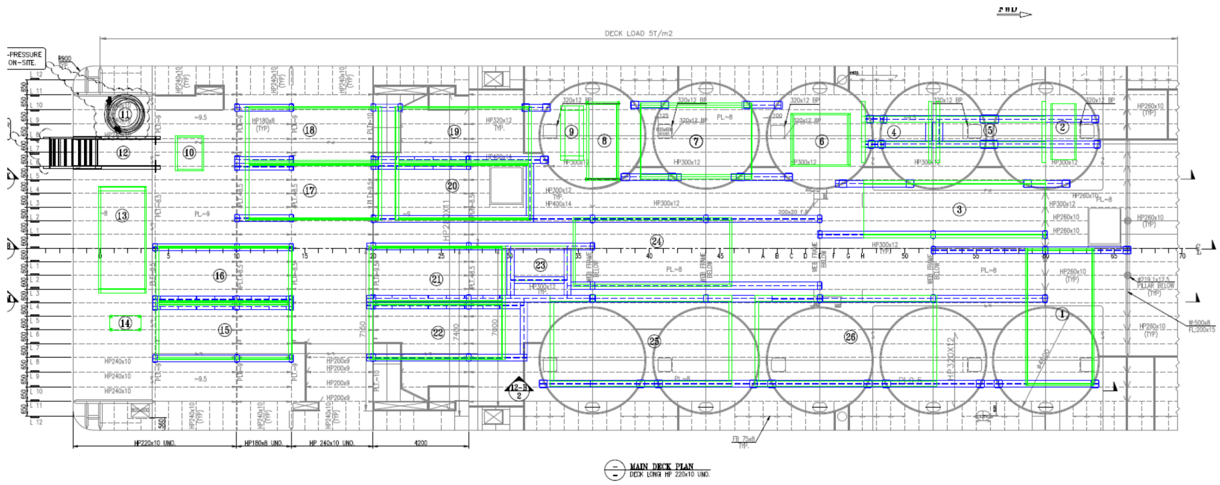

Figure 3—Sea fastening and grillage plan

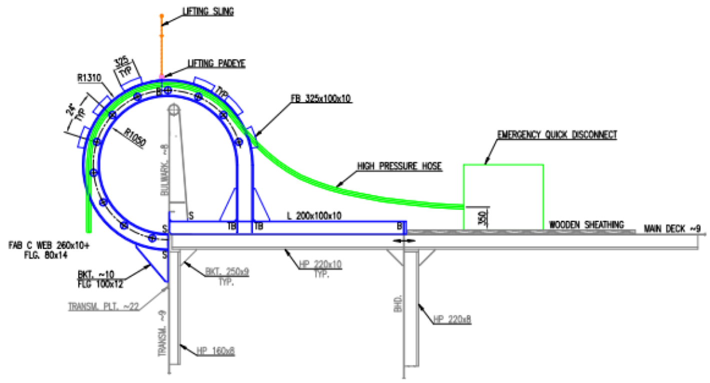

Figure 4—Gooseneck

Hazardous Area Classification and Safety Plan. Identifying and classifying hazardous areas as per IEC 50092-502 rules section 4.5 & Annex E & OSV Chemicals Code, ensures proper safety measures are in place, especially with handling chemicals. The zone 1 included the interior of acid storage tank, tote tanks, interior of piping, mixing tanks and blender. The zone 2 included 1.5 m radius around manhole which is accessible to the tanks area, also the area on the deck within the vertical blender and 3 m radius from the tanks and the vents (Figure 5).

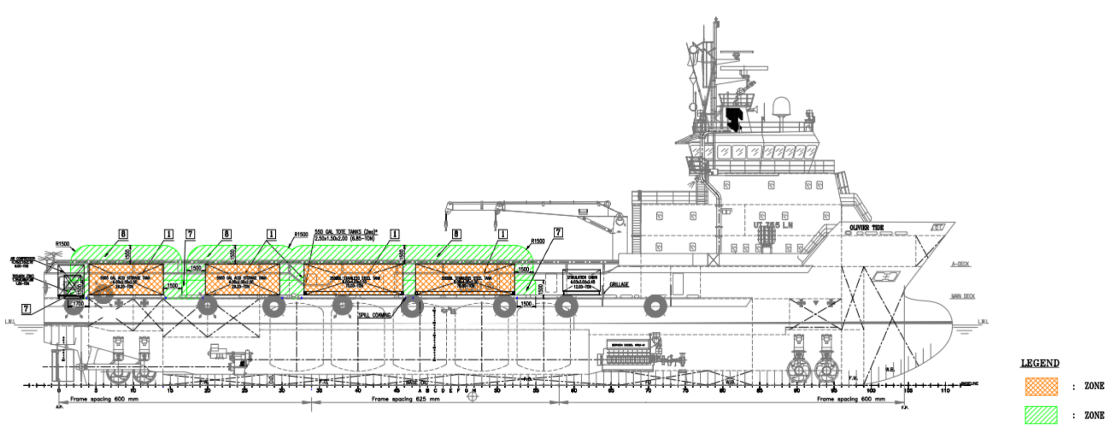

Figure 5—Hazardous Area Plan

An additional fire and safety plan, related to the activities on the deck, was combined with the existing vessel safety plan and implemented during operations.

Acid Resistant Paint. It is essential to protect the vessel and the equipment from corrosion caused by acid with acid resistant paint. Giving the limited time to setup the vessel, getting the appropriate acid-resistant paint was a challenge, and the solution was to source from different nearby country defining a paint layout plan to follow the acid percentage for each area.

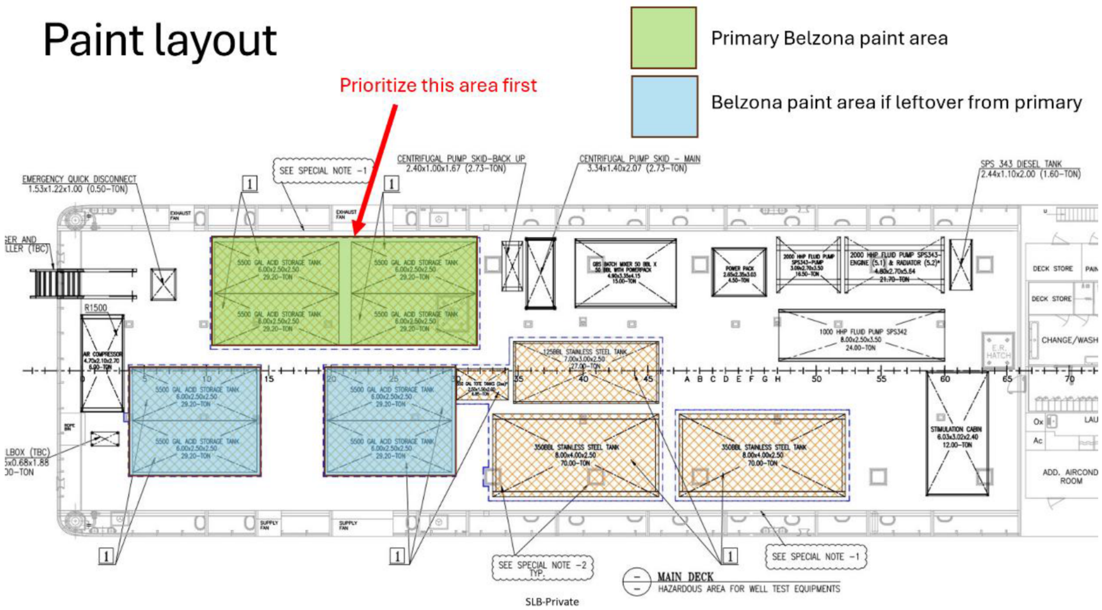

Figure 6—Paint Layout

Review and approval. The entire project was reviewed by the class authority and the vessel flag state. The class authority reviewed and approved the stowage plan, the general arrangement, grillage arrangement, the gooseneck design, the sea fastening design and calculations, the stability calculations, the hazardous area plan and the fire and safety plan, the vent arrangement. These documentation reviews were conducted remotely upon availability of the documents with additional clarifications meeting as needed. The vessel flag authority also reviewed the documentation in addition to the cargo list. Some of the specialized chemicals planned for the operation onboard the vessel had ingredients not listed under the IBC code/OSV Code. After these reviews the flag authority provided exemptions to operate for the specific period to cover the stimulation campaign.

Installation Phase

- The installation started with deck preparation, that involved removal of tuggers, wood sheeting and T -bars, then deck cleaning.
- The prefabricated gooseneck was installed and securely fastened at the stern on the port side of the vessel
- While preparation of grillage was ongoing, the secondary containment around the acid storage and mixing tanks were installed and flood test completed ok.
- Reinforcing the deck: Strengthening the deck with prefabricated grillage was the first step in order to increase the footprint of heavy equipment and avoid exceeding the 5 tons per square meter deck strength.
- Acid resistant paint was applied on the deck, coaming and grillage in the storage and mixing area.

- The acid stimulation equipment was installed on the main deck in 2 steps:
  - A fit trial to ensure the footprint matched the design on paper drawing
  - Then later, the final installation commenced starting with the emergency quick disconnect, the acid mixing tanks, acid mixing blender, the acid storage tanks, the high-pressure pumps, and the c-pumps. the stimulation cabin and the air compressor.
- Sea fastening of the equipment was also completed as per sea fastening drawing and analysis reviewed by class
- Additionally, grating and scaffolding platforms were installed to ensure safe and unobstructed movement for personnel, minimizing trip hazards during operations.
- Safety Protocols: Comprehensive safety protocols were re-evaluated and updated to align with class standards and local regulations for well stimulation operations. This included:
  - Installation of additional firefighting equipment, spill kit, Safety shower, gas detectors.
  - Application of acid-resistant paint in dedicated areas as per class requirement.
  - Then rigorous inspections and obtaining necessary approvals and exception.
- Following the successful installation and fastening of the equipment (Figure 7), a comprehensive sea trial was conducted to ensure the vessel's readiness for offshore operations. The vessel underwent rigorous testing of the high-pressure pumps, acid mixing systems, emergency quick disconnect function test, the rig up of the black eagle hose from vessel to rig deepwater rig. The summary of the key objectives of this operational sea trial were
  - To identify the sailing time from port to rig
  - To identify the rig up time of high-pressure hose with 2 cranes
  - To perform the controlled emergency quick, disconnect with rig team
  - To perform the equipment function test at sea
  - To Install the floater on the black eagle hose and verify the buoyancy o To fit test of black eagle hose landing plate on the hanger installed on the rig.

Figure 7—Installation phase on the deck of PSV

Inspection and Approval

Following the sea trial, a detailed inspection and approval process was undertaken. This included a thorough review by class inspector to verify that all installations met the documentations and regulations. Each component, from the high-pressure pumps to the acid storage tanks, was meticulously examined. The inspection covered both the structural integrity and the operational functionality of the equipment.

The vessel also underwent additional testing for safety features such as the emergency quick disconnect system and the gas detection units. Compliance documents and certifications were reviewed and endorsed by the relevant authorities, ensuring that all modifications and installations were documented and approved.

Final approvals were granted after a series of successful tests, affirming the vessel's readiness for the demanding conditions of offshore acid stimulation operations. This rigorous inspection and approval process ensured that the vessel could operate safely and efficiently, meeting both class standards and local regulatory requirements.

Observations and Results

The observations from the comprehensive sea trial were instrumental in assessing the vessel's overall performance and readiness. The high-pressure pumping setup demonstrated reliability and efficiency, consistently meeting operational specifications under varying conditions. The acid mixing plan performed seamlessly, ensuring precise and safe handling of chemicals for well stimulation.

The emergency quick disconnect system was tested rigorously, proving its efficacy in rapid-response scenarios. The gas detection units were also tested thoroughly, confirming their accuracy and responsiveness in detecting potential hazards. All safety equipment, including firefighting systems and safety showers, were operational and met the required standards.

The performance metrics gathered from these trials indicated that the fit for purpose stimulation vessel (Figure 8) was not only compliant with class standards but also capable of exceeding operational expectations. The results were a testament to the meticulous planning, execution, and teamwork that went into the project. The vessel was now ready to undertake offshore acid stimulation operations with confidence, ensuring both safety and productivity.

Figure 8—Stimulation vessel in operation

Stimulation project performance

Stimulation from PSV have been executed safer and faster compared to drilling rig solution and analysis of the operation clearly certifies the benefit of this approach.

Due to the time to prepare the PSV, few stimulation jobs have been executed from Drilling Ship deck. Due to limited space, Operator had the possibility to spot only 1000 bbl per batch and then stop the operation for refill of acid tanks and then restart the operation with a 24 hr time needing for mixing operation.

With the use of stimulation vessel, Operator executed all the batch of 2000 bbl per well in one run, minimizing the time and risk associated to CT in well.

Comparing the two scenario, PSV solution allowed a reduction up to 2 days per well, impacting roughly with $2 million saving for each job, with additional benefit in terms of logistics and flexibility, since drilling ship deck was free to accommodate other equipment needed for the job.

As matter of example, during the campaign, an unexpected well problem forced the team to postpone the acid treatment: due to location of Acid tank in the vessel and space available on drilling ship, Operator decided to modify the plan and execute recovery job on the well without any additional impact on the new plan for acid job re-phasing.

Moreover, the vessel was also used as liquid cargo transport for rig activities. This dual functionality of the PSV provided additional logistical benefits, as it reduced the need for separate transportation arrangements and allowed for more efficient use of resources. The ability to transport liquid cargo alongside conducting stimulation operations exemplified the versatility and practicality of utilizing such vessels in offshore projects.

Conclusion

One of the most significant examples of success in offshore acid stimulation operations using coiled tubing (CT) was made possible through the conversion of a Platform Supply Vessel (PSV) into a dedicated Stimulation Vessel for deepwater operations. The vessel, originally designed solely for transporting materials and supplies to oil platforms, underwent a structural and system overhaul that expanded its technical capabilities. As a result, it became fully suited to house high-pressure acid pumping systems, treatment modules, specialized tanks for corrosive fluids, and advanced safety systems compliant with international regulations for handling hazardous substances offshore.

The project was initiated with the goal of optimizing the time-to-market of acid stimulation jobs in offshore wells. Traditionally, these activities were carried out using equipment on drilling ship or dedicated stimulation vessel (as built), which required costly mobilization and reconfiguration for non-drilling operations or, in the case study, can occur in not availability of the resource. This not only increased expenses but also had a negative impact on operational timelines, due to the time involved for rig-up and the logistical challenges associated with managing materials, specialist equipment, and personnel.

With the commissioning of the converted PSV in Stimulation Vessel, a significant improvement in operational performance was observed: the complete operational cycle (mobilization, positioning, execution of the treatment, demobilization) showed an average reduction of two days per intervention compared to equivalent operations performed from a drilling ship. This time advantage was achieved through several synergistic factors. The converted PSV, being specifically designed for these types of interventions, can keep key acid stimulation permanently on board, eliminating the need for frequent module installations and removals.

In addition to logistical and operational benefits, the increased storage capacity and simultaneous management of different fluids enabled treatments without the need to refill or return to port, thereby increasing the overall efficiency of the stimulation campaign. Process automation, remote monitoring of pumping parameters, and the integration of advanced safety and emergency management systems further contributed to a significant reduction in Non-Productive Time (NPT) and to an increase in safety for onboard personnel.

Analysis of the data collected during the campaign of 4 interventions showed an average reduction of 2 days per job compared to historical data for the same procedures conducted via drilling ship, translating into estimated economic savings average $2 million per operation, depending on asset type and market conditions. These results not only confirm the strategic value of the PSV-to-Stimulation Vessel solution but also strengthen the operator's competitive position in offshore deepwater stimulation campaigns, providing a replicable model for future conversion initiatives.

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